UNIVERSITÀ DEGLI STUDI DI MILANO BICOCCA

Sviluppo mobile con supporto di LLM

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Obiettivo

- Diffusione di strumenti LLM in tempi recenti
- ChatGPT, Gemini, Copilot...
- Quanto supporto può fornire uno strumento come ChatGPT agli sviluppatori?

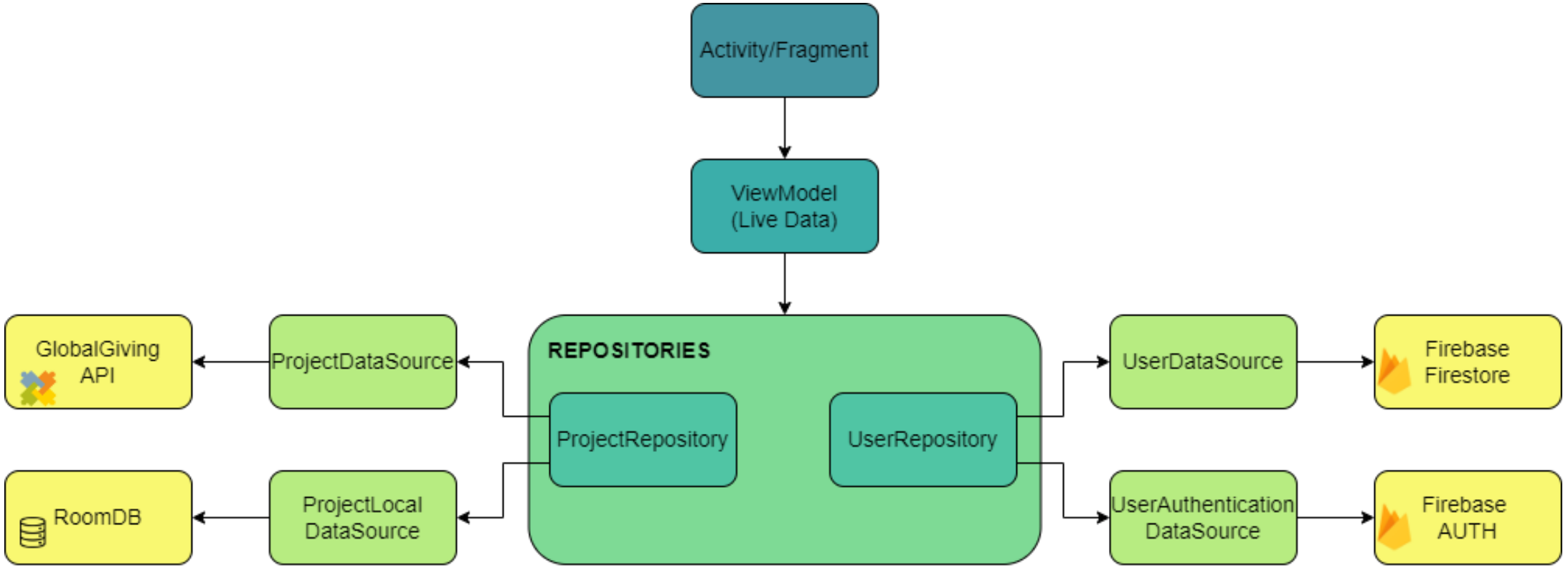
Metodo di verifica

- Sviluppo di un’applicazione nell’ambito sociale
- Annotazione delle richieste a ChatGPT utilizzando un diario nelle seguenti fasi:
  - Requisiti
  - Progetto architetturale
  - Sviluppo
  - Testing

Requisiti funzionali

- Registrazione
- Login
- Logout
- Visualizzazione progetti
- Visualizzazione dettagli progetto
- Filtraggio progetti
- Donazione

Architettura



Testing

Framework utilizzati:

- Mockito
- Espresso

Test effettuati:

- Repositories
- ViewModels
- UI

```
no usages  Marconapoo
@Test
public void testOnProjectsLoaded() {
    ProjectsApiResponse projectsApiResponse = mock(ProjectsApiResponse.class);
    ProjectsApiResponse.Search search = mock(ProjectsApiResponse.Search.class);
    ProjectsApiResponse.Response response = mock(ProjectsApiResponse.Response.class);
    ProjectsApiResponse.ProjectData projectData = mock(ProjectsApiResponse.ProjectData.class);
    List<Project> projectList = new ArrayList<>();
    Project project = mock(Project.class);

    projectList.add(project);  // Add a mock project to the list

    // Mock nested calls to avoid NullPointerException
    when(projectsApiResponse.getSearch()).thenReturn(search);
    when(search.getResponse()).thenReturn(response);
    when(response.getProjectData()).thenReturn(projectData);
    when(projectData.getProjectList()).thenReturn(projectList);
    when(project.getTitle()).thenReturn( value: "Project Title");

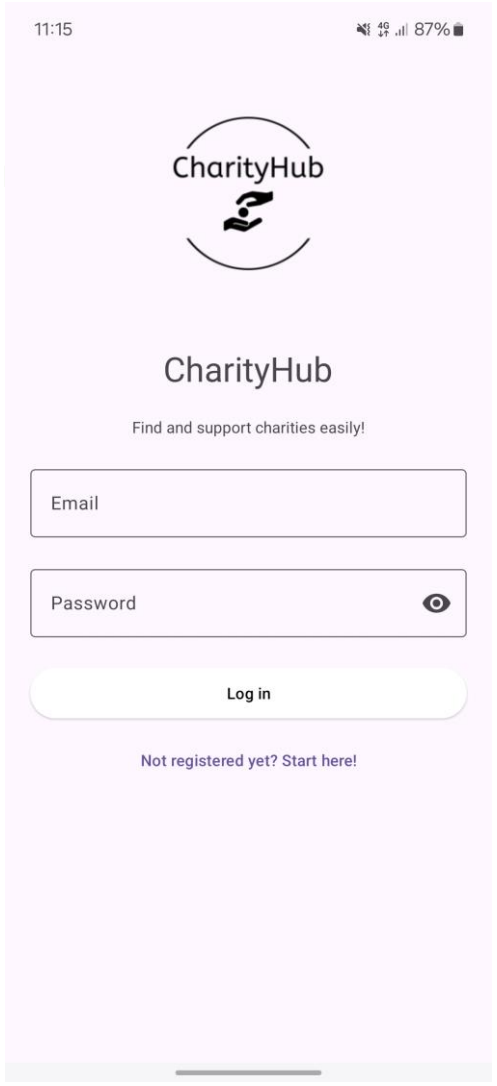
    projectRepository.onProjectsLoaded(projectsApiResponse);

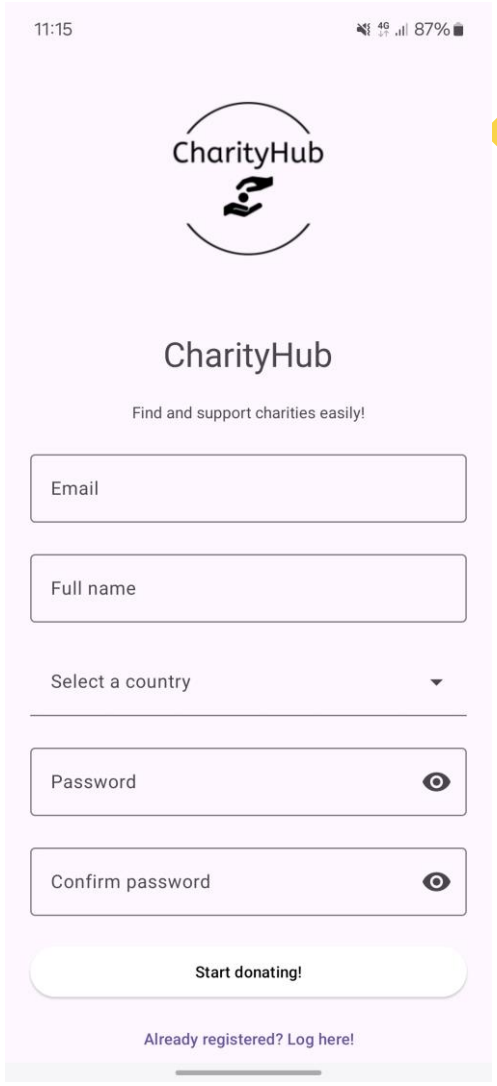
    verify(projectLocalDataSource).insertProjects(projectsApiResponse);
}

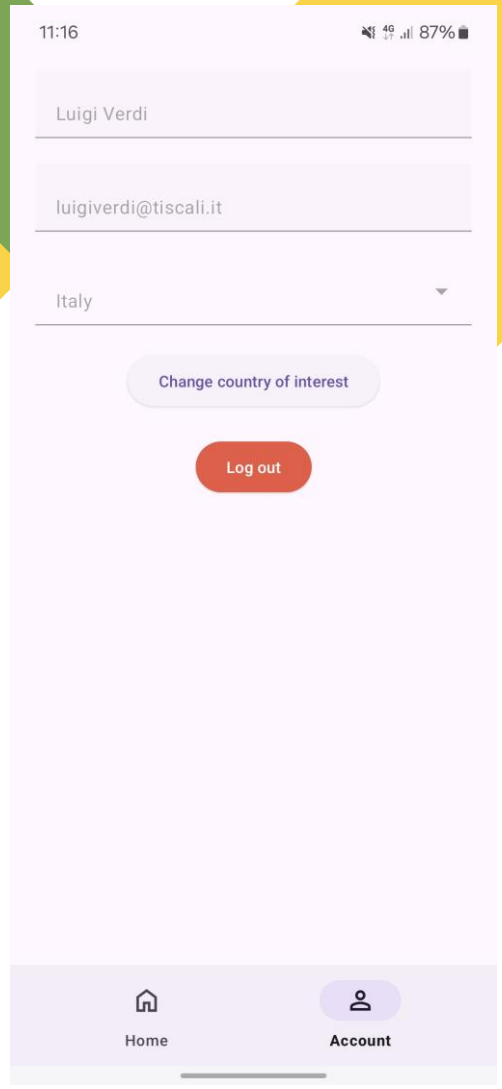
no usages  Marconapoo
@Test
public void testOnFailureFromRemote() {
    String error = "error message";
    projectRepository.onFailureFromRemote(error);

    ArgumentCaptor<Result.Error> captor = ArgumentCaptor.forClass(Result.Error.class);
    verify(projectsLiveData).postValue(captor.capture());
    assertEquals(error, captor.getValue().getErrorMessage());
}
```

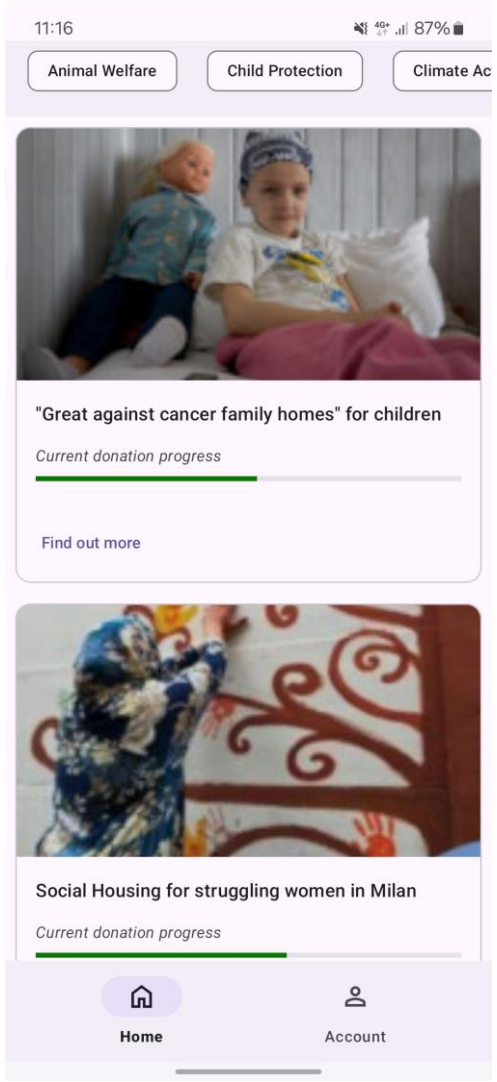
User Interface (1)

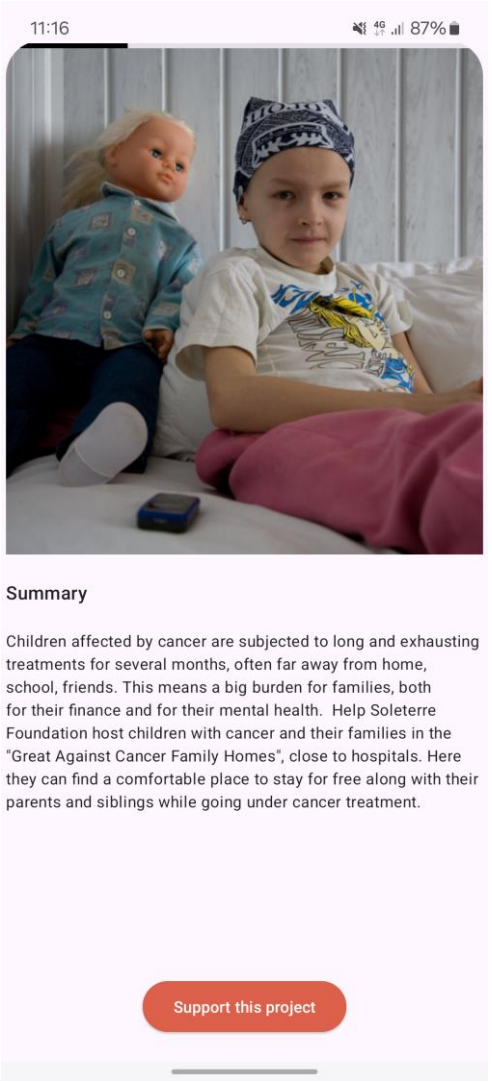


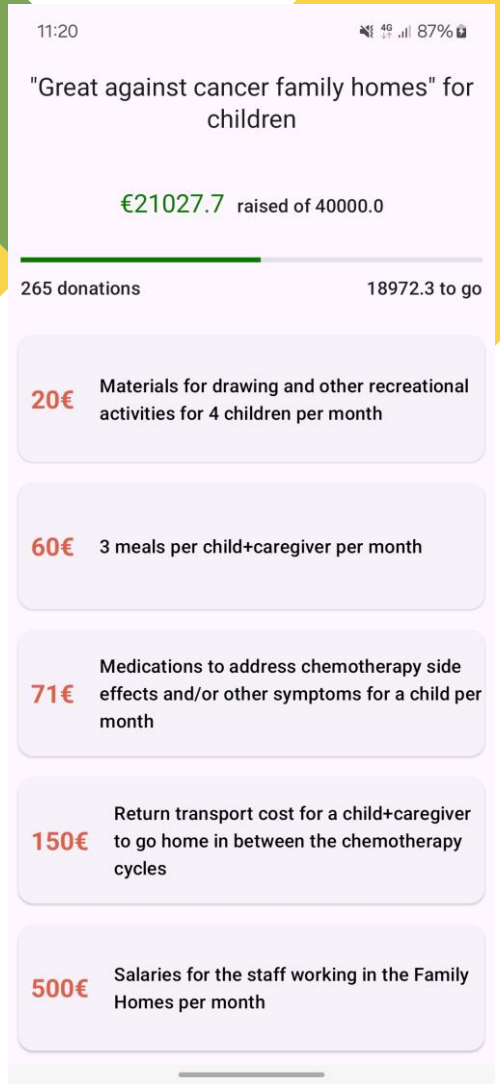




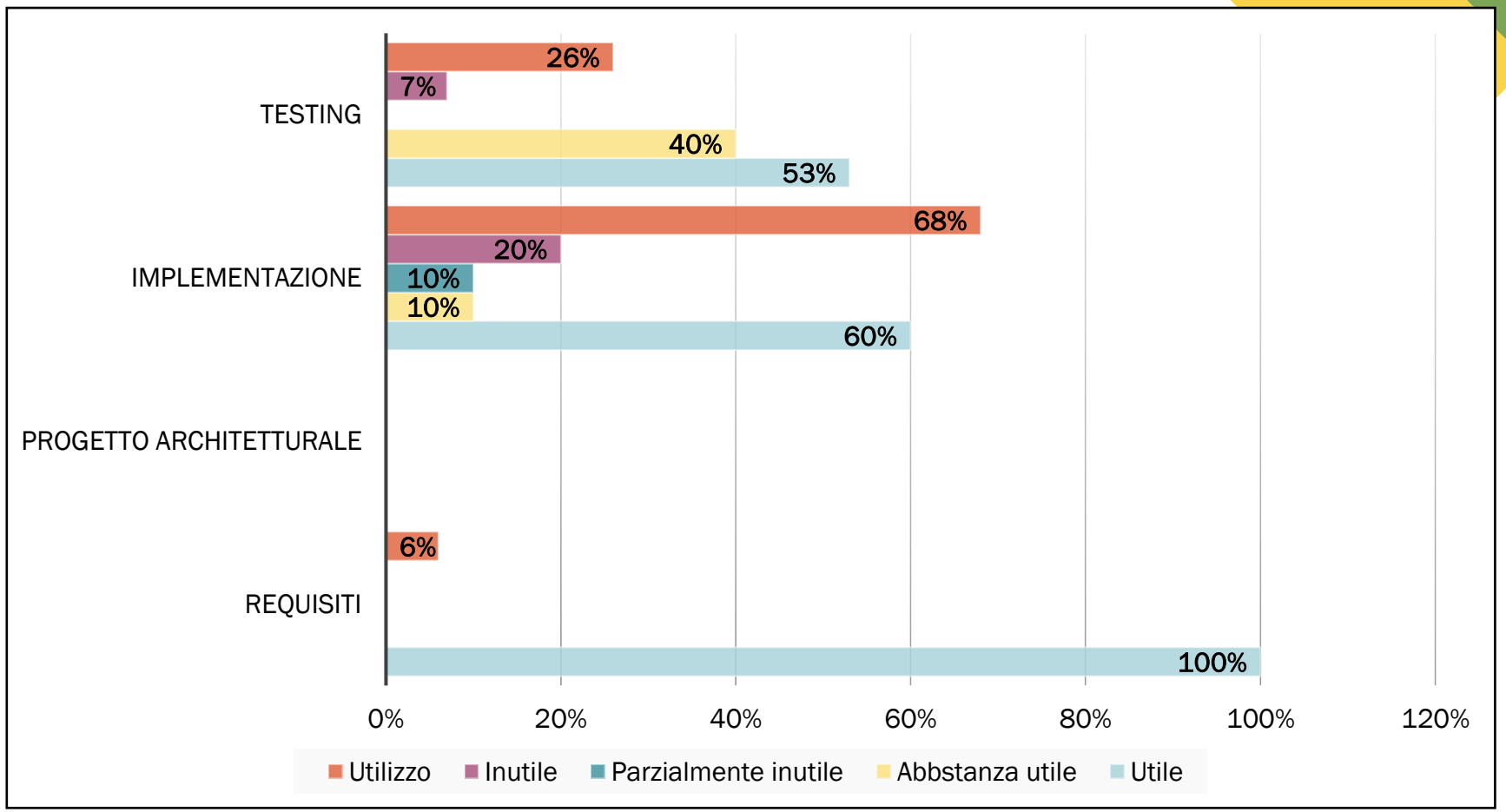
User Interface (2)







Utilizzo di ChatGPT



Conclusioni

- LLM per programmatori esperti e principianti
- Risorsa aggiuntiva a quelle esistenti
- ChatGPT è uno strumento e non un sostituto
- L’importanza delle conoscenze pregresse